1171: Maximo AI Service and IBM Bob

Hands-on lab guide

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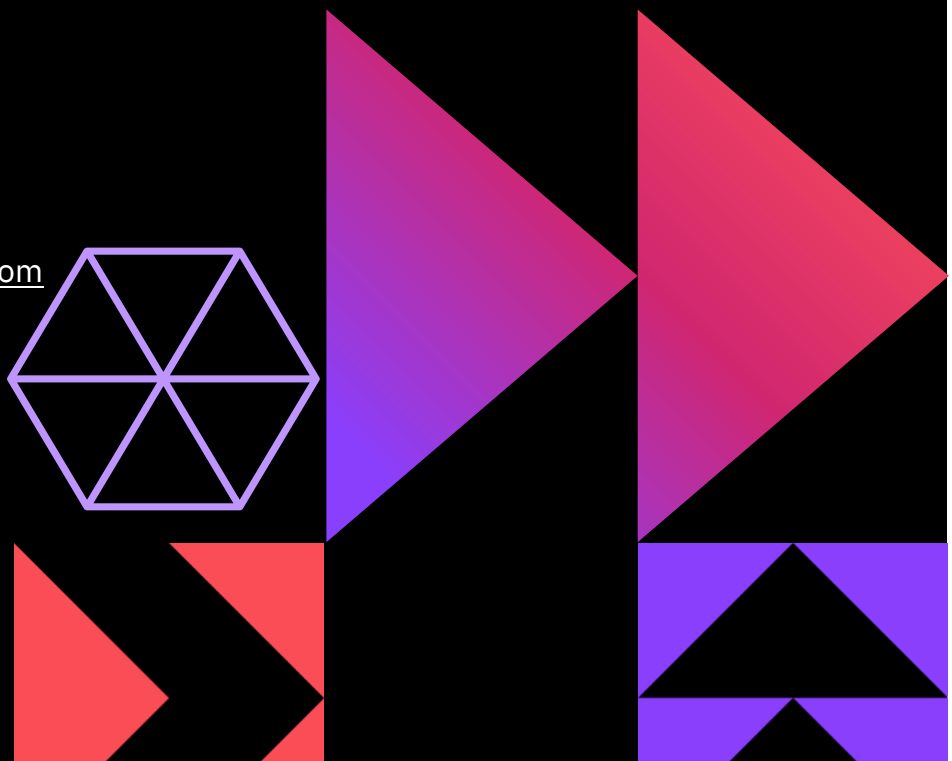


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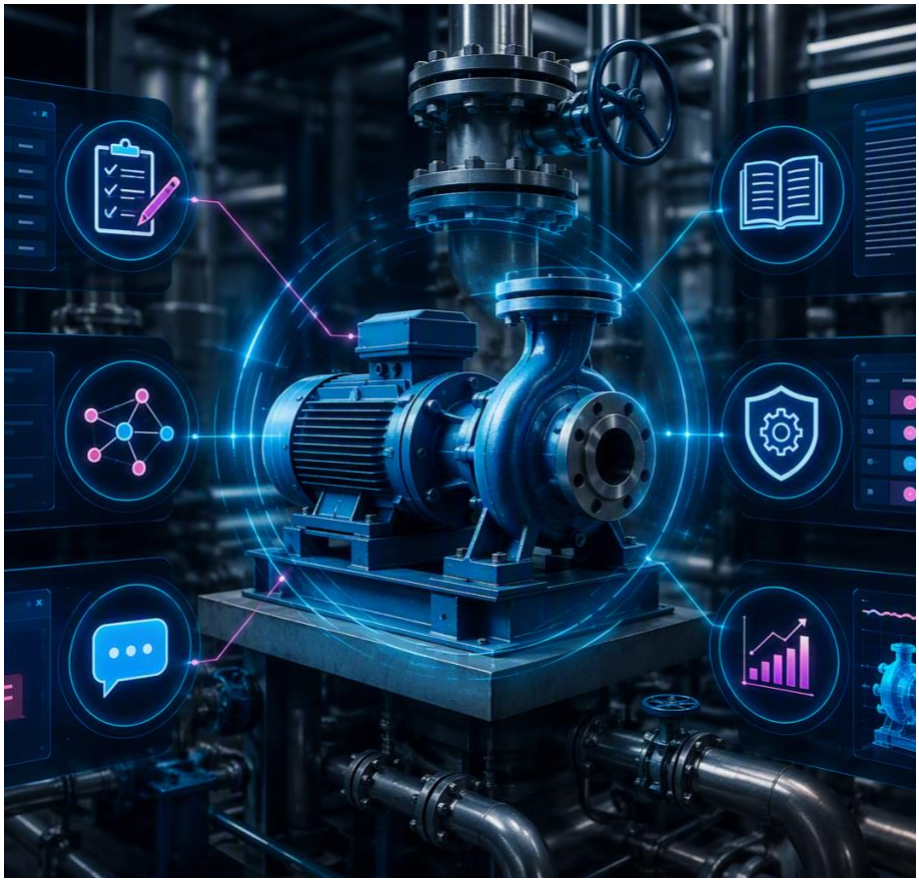
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Lab Reference Materials

https://ibm.biz/1171-links

1 Introduction

This hands-on workshop introduces seven AI-powered capabilities in IBM Maximo Application Suite Manage 9.2. The exercises use pump assets, operating locations, work-order history, meter data, and alerts that have been preloaded for the BEDFORD site. Each exercise focuses on a practical maintenance or reliability workflow and allows participants to use the AI capability directly in a live environment.



1.1 About this hands-on lab

By the end of this workshop, you will have used AI in Maximo to classify work, reuse historical maintenance experience, query operational data, search product documentation, build a reliability strategy, assess asset condition, and transform an operational alert into an actionable maintenance response.

#	Use case	AI tool	Time
1	Recommend a problem code for a new work order	Problem Code Classification (`pcc`)	15 min
2	Find similar historical work orders	Work Order Similarity (`similarity`)	15 min
3	Query Maximo data with natural language	Chat Assistant (`nl2oslc`)	20 min
4	Ask questions about Maximo product documentation	IBM Docs Assistant (`docsearch`)	10 min
5	Generate a reliability strategy with AI	FMEA Strategy Assistant (`fmea`)	20 min
6	Assess asset health and get AI condition insights	Asset Insights Generator (`insightsgenerator`)	20 min
7	Turn an operational alert into an actionable maintenance response	Alert Insights	20 min

1.2 Workshop environment

Item	Value
Environment	IBM Maximo Application Suite Manage 9.2 with AI Service
Site	BEDFORD
Organization	EAGLENA
URL	TBA
Credentials	TBA

1.3 What has been loaded

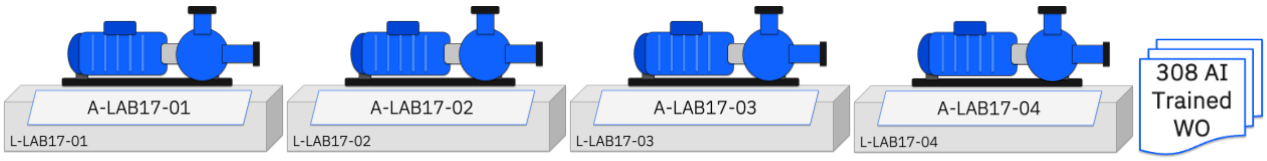
Before you begin, review the data that has been loaded into the workshop environment.

Object	Details
Locations	`L-LAB0X-01` through `L-LAB0X-04`, four pump-house operating locations in BEDFORD
Assets	`A-LAB0X-01` through `A-LAB0X-04`, four submersible pump assets with serial numbers, install dates, meters, and failure class `PUMPS`
Work orders	77 completed work orders per asset, 308 total, covering problem codes **STOPPED**, **LOWPRES**, **LEAK**, and **LOWVOL** across CM and EM work types; all are flagged for AI training

From the left-side navigation menu, you can access and review all data that has been preloaded for this lab.

In the filter navigation field enter “Assets” or “Work order Tracking” to review preloaded data.

Visual example of loaded data:



2 Recommend a Problem Code for a new Work Order

2.1 What you will learn

How the **Problem Code Classification** AI tool reads a work-order description and suggests the most likely problem code, saving time and improving data quality.

2.2 Background

The `pcc` model has been trained on 308 completed work orders. Each training work order includes a description, failure class `PUMPS`, and an accurate problem code. When you create a work order, the AI analyzes the description and returns the most likely problem code with a confidence score.

2.3 Steps

Anywhere you see a mention of A-LABXX-, assume that you must replace XX with your respective lab ID.

2.3.1 Navigate to Work Orders

1. From the MAS Manage home page, search for **Work Orders**. Or navigate the menu **Work Orders** – **Work Orders**.
2. From the top-right hand corner, click **Create work order** **+** button.

2.3.2 Describe the problem

1. Pick one of the below mentioned descriptions and enter it in the **Description** field. Add your `LABXX` username to the end of the description.
2. Set **Work Type** to `CM` or `EM`.
3. Set **Asset** to `A-LABXX-01`.
4. Click **Create** and write down the work-order number.

Note: Location and Failure class will be picked up automatically. DO NOT define Problem Code.

Write down the work-order number after you click Save in step 5.

```
Pump tripped offline and will not restart - motor appears seized LABXX
```

```
Discharge pressure has dropped well below normal operating range LABXX
```

```
Visible fluid leak found at the discharge flange connection LABXX
```

```
Flow rate at storage tank dropped significantly overnight LABXX
```

2.3.3 Review the AI recommendation

1. Wait a minute or two. Open the saved work order and review the **Problem code** field. The AI should suggest `STOPPED`, `LOWPRES`, `LEAK`, or `LOWVOL`. Review the confidence score.
2. Click the **AI icon** to review the model description.
3. **Edit** the work order. Under Problem Code field, click Recommendation +2 more. You have the option to accept the suggestion or edit the description and regenerate.

If time permits, create few more work orders and observe.

2.3.4 Accept or override

1. If the suggested code is correct, leave it as is or accept it.
2. To use a different code, enter it directly in the **Problem code** field. The AI suggestion is advisory.

3 Find Similar Historical Work Orders

3.1 What you will learn

How the **Work Order Similarity** AI tool finds semantically related work orders so technicians can reuse resolution steps from past repairs.

3.2 Background

The `similarity` tool uses an embedding model, with no training required, to rank eligible work orders by similarity. Results are filtered by site.

3.3 Steps

3.3.1 Open an existing work order

1. Go to the **Work Orders** application.
2. Filter by **Asset** = `A-LAB0X-01`.
3. Open the work order you created earlier.

3.3.2 View similar work orders

1. Open the **Actions** menu.
2. Select **View similar work orders**.
3. Review the side panel containing work orders ranked by similarity.

Click the Regenerate button if the similar work orders do not appear the first time.

3.3.3 Explore the results

1. Open a similar work order and compare its **Description** with the original.
2. Review **Actual work** or **Failure remarks** and decide whether the resolution is transferable.
3. Click the **AI icon** to view model metadata.

3.3.4 Use the results

1. Select two or three similar work orders.
2. Open the **Change status** batch action to see how results can support bulk operations.

3.3.5 Try another example

1. Repeat the exercise.

4 Generate a Reliability Strategy with AI

4.1 What you will learn

How the **FMEA Strategy Assistant** (`fmea`) accelerates Failure Mode and Effects Analysis by proposing asset boundaries and failure modes.

4.2 Background

The `fmea` tool uses a generative AI model to propose asset boundaries and failure modes from its knowledge of industrial equipment. No training data is required. Review, accept, or refine all suggestions before saving.

4.3 Steps

4.3.1 Navigate to Reliability Strategies

1. Open **Menu > Assets > Reliability Strategies**.

4.3.2 Create a strategy

1. Click **Custom strategies**. Click **Create strategy**.
2. Enter `PUMPSTRATEGY##`, replacing `##` with your user number.
3. Enter `Reliability strategy for submersible pump assets` as the description.
4. Set **Asset classification** to the pump class, such as `PUMP`.

4.3.3 Generate a boundary

1. Scroll down to the Boundary section. Click **Generate recommendation AI**

3. Review the proposed boundary and edit it if needed. Click **Create**. Save any comment and click Create. The strategy will be saved as a **Draft**.

4.3.4 Generate failure modes

1. Open the draft strategy. Click to open **Failure modes**.
2. Click and select the hierarchy **icon** from the right.
3. Click **Generate recommendation** link.
4. Select relevant failure modes and click **Accept** or **Add selected**.

4.3.5 Save the strategy

1. Click **Save**.
2. Optionally change the status from **Draft** to **Completed**.

5 Assess Asset Health and Get AI Condition Insights

5.1 What you will learn

How the **Asset Insights Generator** (`insightsgenerator`) combines work-order history, meter readings, asset details, and the Maximo Health score into a condition summary and prioritized recommendations.

5.2 Background

The model uses predefined Maximo data sources to generate a condition summary, specific insights with confidence levels, and recommended actions. No training is required.

5.3 Steps

5.3.1 Navigate to Assets & Locations

1. Open **Assets & Locations**.
2. Locate asset `A-LABXX-01` at site **BEDFORD** and open the asset card.

5.3.2 Locate the Insights card

1. Open the **Insights** card or tab.
2. An empty state is expected if insights have not been generated.

5.3.3 Generate insights

1. Click **Generate** or **Generate insights**.
2. Allow 2 to 5 minutes for processing.
3. Stay on the page or click **Refresh** when results are ready.

5.3.4 Review the condition summary

1. Read the **Condition summary**.
2. Note references to pump type, location, failure patterns, and Health score.

5.3.5 Explore insights

1. Review observations and confidence badges.
2. Expand an insight to read its reasoning.

5.3.6 Review recommendations

1. Review the highest-priority inspection, PM, or data-quality action.

5.3.7 Compare assets

1. Open `A-LABXX-02` and generate insights.
2. Compare the condition summaries and recommendations with `A-LABXX-01`.

5.3.8 Regenerate

1. Return to `A-LABXX-01` and click **Regenerate**.
2. Observe whether text or confidence changes.

5.4 Tips for better insights

Assets with more CM and EM work-order history generally produce richer insights.

A configured, current Health score improves the condition summary.

If the card shows **Not enough data**, confirm that completed work orders exist and the AI configuration is active.

6 Transform an Operational Alert into Actionable Maintenance

6.1 What you will learn

How **Alert Insights** analyzes an operational alert, explains the likely failure, recommends maintenance actions, and helps create a work order with the analysis included.

6.2 Background

Operational alerts indicate that something happened but may not explain why or what maintenance response is appropriate. In this exercise, you will analyze a high-vibration alert for a pump and create a work order from the reviewed insight.

6.3 Steps

6.3.1 Navigate to Assets & Locations

1. From the menu, select **Assets** and then **Assets & Locations**.
2. Confirm that the **Assets** tab is selected.

6.3.2 Open the asset

1. Locate `A-LAB01-02` and open the asset details.

6.3.3 Open Alerts and meters

1. Click **Alerts and meters**.
2. Open the first alert link. This opens a Ticket. Drop-down Actions menu and create Work Order. Review the alert summary by priority.
3. Open `Discharge pressure loss-suspected impeller seizure`.

6.3.4 Review the alert

1. Review status and summary information.
2. Confirm the alert is associated with `A-LAB01-02`.
3. Scroll to **Insights**.

6.3.5 Open Alert Insights

1. Click **View more**.
2. Review **Summary**, **Failure analysis**, and **Recommendations**.
3. Note how the analysis explains that discharge pressure has dropped, which is commonly caused by the impeller becoming seized and unable to rotate.

6.3.6 Review the explanation

1. Click the **AI information icon**.
2. Review the explanation and close the panel.

6.3.7 Create a work order

1. Return to the alert.
2. Select **Actions > Create work order**.
3. Review the prepopulated **Summary** and **Description**.
4. Confirm that alert context, AI summary, failure analysis, and recommendations are included.
5. Complete required fields and click **Create**.

7 Query Maximo Data with Natural Language

7.1 What you will learn

How the **Chat Assistant** (`nl2oslc`) answers plain-English questions about work orders, assets, and locations without requiring OSLC queries or SQL.

7.2 Background

The `nl2oslc` model translates natural language into an OSLC query, runs the query against Maximo, and returns matching records. The assistant respects security permissions and does not create or modify records.

7.3 Steps

7.3.1 Open the AI Assistant

1. Click the **chat bubble** in the top-right navigation bar.
2. The Chat Assistant panel opens on the right.

7.3.2 Ask work-order questions

1. Enter the work-order prompts below one at a time.
2. Review the returned columns and record count.
3. Click a work-order number to open the record.

```
Show me all open work orders for asset A-LABXX-01
```

```
Show work orders with problem code STOPPED at site BEDFORD
```

```
How many work orders were completed in the last 30 days?
```

```
Show me emergency work orders for the PUMPS failure class
```

```
Show me all assets at location L-LABXX-01
```

```
Show assets with failure class PUMPS at site BEDFORD
```

```
Which assets are missing an installation date?
```

```
What is the total number of work orders per problem code for asset A-LABXX-02?
```

```
How many emergency work orders exist for each asset?
```

```
Create a work order for the pump at location L-LABXX-03
```

7.3.3 Ask asset questions

1. Enter the asset prompts below one at a time.

7.3.4 Try aggregations

1. Enter the aggregation prompts below.

7.3.5 Observe limitations

1. Ask the assistant to create a work order. The assistant should decline because it does not perform write operations.

8 Ask Questions About Maximo Product Documentation

8.1 What you will learn

How the **IBM Docs Assistant** (`docsearch`) returns answers grounded in official IBM product documentation without leaving MAS Manage.

8.2 Background

The `docsearch` tool performs a semantic search of IBM documentation for MAS Manage and returns a synthesized answer with source references.

8.3 Steps

8.3.1 Open the AI Assistant

1. Click the **chat bubble** to open the assistant panel used in the previous exercise.

8.3.2 Ask product questions

1. Enter each prompt below.
2. Read the synthesized response.
3. Review and open the source references to verify the answer.

```
What is the purpose of the failure code field on a work order?
```

```
How do I configure a preventive maintenance schedule in Maximo?
```

```
What security groups are required to use reliability strategies?
```

```
What is the difference between a work order and a service request?
```

```
How does the AITRAINJOB cron task work?
```

```
How does the problem code classification AI tool work?
```

```
What training data is required for the Work Order Similarity feature?
```

```
What permissions does a user need to view similar work orders?
```

8.3.3 3. Ask about AI capabilities

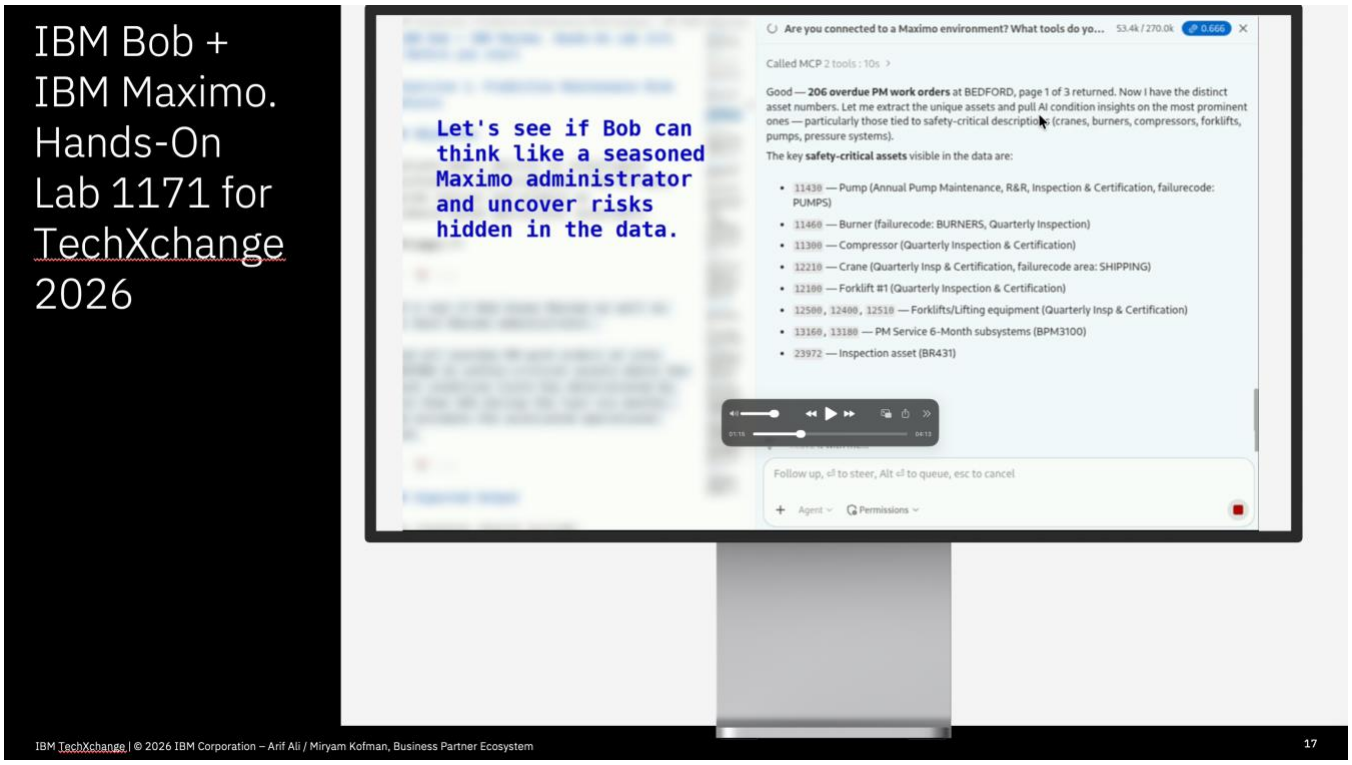
1. Ask the assistant about the AI tools used in the workshop.

8.3.4 4. Try an out-of-scope question

1. Ask `What is the capital of France?` and observe that the assistant remains focused on IBM product documentation.

Hands-on with IBM Bob connected to Maximo using MCP. Refer to Instructors for the next steps

Preview video



https://ibm.box.com/s/5k79je8ofbakc2j2wqd1u4g5svct5xff

9 Quick reference: Assets and work orders

Asset	Location	Failure class	Problem codes in history
A-LABXX-01	L-LABXX-01	PUMPS	STOPPED, LOWPRES, LEAK, LOWVOL
A-LABXX-02	L-LABXX-01	PUMPS	STOPPED, LOWPRES, LEAK, LOWVOL
A-LABXX-03	L-LABXX-01	PUMPS	STOPPED, LOWPRES, LEAK, LOWVOL
A-LABXX-04	L-LABXX-01	PUMPS	STOPPED, LOWPRES, LEAK, LOWVOL

Each asset has approximately:

52 CM work orders with problem code STOPPED

52 CM work orders with problem code LOWPRES

52 CM work orders with problem code LEAK

52 CM work orders with problem code LOWVOL

8 EM work orders spread across all four problem codes

10 AI Tool Summary

Tool	MAS configuration	Training	Where visible
Problem Code Classification	WOPROBLEMCODE	Yes, completed	Problem code field on Work Order
Work Order Similarity	WOSIMILARITY	No	Actions > View similar work orders
Chat Assistant	ASSISTANT	No	Chat bubble in top navigation
IBM Docs Assistant	DOCSEARCH	No	Chat bubble, documentation mode
FMEA Strategy Assistant	RSSTRATEGYASSISTANT	No	Generate recommendation in Reliability Strategies
Asset Insights Generator	INSIGHTSGENERATOR	No	Insights card on asset or location details
Alert Insights	Alert Insights	No	Alert > Insights > View insights; Actions > Create work order

11 Getting help and troubleshooting

11.1 Common troubleshooting tips

Symptom	Likely cause	What to try
No AI problem-code suggestion	Failure class or asset is missing	Set **Failure class = PUMPS**, assign an asset, and save.
View similar work orders is missing	Permission or inactive configuration	Ask the instructor to verify `WOSIMILARITY`.
Assistant returns an empty result	Prompt is vague or lacks context	Include site, asset, and date-range values.
FMEA recommendation is unavailable	Configuration or model is not ready	Ask the instructor to verify `RSSTRATEGYASSISTANT`.
Strategy creation is unavailable	Required strategy access is missing	Ask the instructor to verify access.
Insights shows Not enough data	Insufficient asset history	Confirm completed work orders and regenerate.
Insights generation does not complete	AI service or connectivity issue	Refresh and regenerate; notify the instructor if the issue persists.
Alert Insights is not visible	No insight content or feature configuration issue	Open the specified high-vibration alert and ask the instructor to verify configuration.
Create work order is unavailable	Insufficient work-order permission	Ask the instructor to verify access.
Work-order description is not prepopulated	Alert insight was not carried into the flow	Reopen View insights and start Create work order again.

12 Lab reference materials

Use IBM product documentation and the IBM TechXchange 2026 lab-download site for additional learning and post-event reference material.

Workshop prepared for MAS Manage 9.2 AI Service, BEDFORD / EAGLENA demo-data environment.